Language-like Predictability and Structural Divergence in Voynich Tokens: A Comparative Statistical Study

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Data source: Zandbergen-Landini EVA transcription (ZL3b)

AI Use Disclosure

AI-based tools (Claude, ChatGPT, Gemini) were used to assist with code implementation, figure generation, and manuscript editing. Research design, hypothesis formulation, analytical decisions, and interpretation were conducted by the author.

Abstract

The Voynich Manuscript has long resisted definitive classification, with competing hypotheses proposing natural language encoding, artificial construction, or meaningless text. Rather than attempting direct decipherment, this study examines the statistical structure of Voynich tokens using a reproducible, quantitative framework. A set of metrics based on prefix-conditioned predictability, entropy decay, and ambiguity persistence is introduced, and Voynich text is compared against multiple baselines, including natural languages (English and Latin), synthetic n-gram text, and structured data. To ensure robustness, size-matched sampling and permutation-based significance testing are applied. The results indicate that Voynich tokens exhibit strong statistical similarity to natural language across both analytic (English) and inflectional (Latin) systems ($\rho \approx 0.95$, $p < 0.01$). However, structural analysis reveals minimal overlap in prefix-family organization and highly constrained suffix distributions. Furthermore, null models fail to reproduce the observed persistence of ambiguity. These findings suggest that Voynich text represents a structured symbolic system combining language-like predictability with distinct generative constraints, rather than a direct encoding of known natural language or a trivial artificial construct.

Keywords: *Voynich Manuscript, entropy, predictability, statistical linguistics, token structure, null models*

1. Introduction

The Voynich Manuscript remains one of the most enduring unsolved problems in historical linguistics and cryptography. Since its discovery, numerous interpretations have been proposed, ranging from natural language encipherment to elaborate hoaxes. Despite extensive efforts, no consensus has been reached regarding the nature of its underlying system. Previous research has demonstrated that the Voynich text is not random, exhibiting statistical regularities such as Zipf-like distributions and positional dependencies. However, these observations alone do not clarify whether the text is fundamentally linguistic, symbolic, or artificially generated.

This study approaches the problem from a statistical perspective. Rather than attempting to decode the text, it evaluates the behavior of Voynich tokens relative to known data types. The objective is to determine whether Voynich text aligns more closely with natural language, random text, or structured artificial systems.

2. Research Questions

RQ1. Does Voynich text exhibit statistical similarity to natural language across different language types?

RQ2. Are the observed statistical properties reproducible by simple generative or random models?

RQ3. Does Voynich text share structural organization with natural language at the level of token families and suffix distributions?

RQ4. Can language-like predictability coexist with non-linguistic structural constraints in the same system?

3. Related Work

Previous research on the Voynich Manuscript has primarily focused on determining whether the text is meaningful or random. Early statistical analyses identified non-random distributions, including Zipf-like word frequency patterns consistent with natural language behavior (Bennett, 1976; Landini, 2001). Subsequent studies examined positional dependencies, demonstrating that token structure varies systematically across line and paragraph boundaries (Montemurro & Zanette, 2013). More recent work has applied information-theoretic approaches, including entropy and mutual information, to characterize the internal structure of the text.

In parallel, generative hypotheses have been proposed, ranging from stochastic text generation to algorithmic hoax models (Rugg, 2004). However, such models generally fail to reproduce higher-order dependencies observed in the manuscript.

More recent work has extended statistical analysis to positional constraints, demonstrating that specific token classes are systematically bound to line-initial, paragraph-initial, and line-final positions under multiple permutation-based null models (Chang, 2026a). The present study complements this line of inquiry by shifting focus from positional constraints to intra-token predictability structure, enabling a direct comparison between Voynich and known language systems. Together, these two approaches constitute a converging body of quantitative evidence for the structured, non-random nature of the manuscript.

4. Methods

4.1 Data

The analysis is based on the ZL3b transcription of the Voynich Manuscript, selected for its explicit handling of ligatures and token boundaries. Baseline datasets include English (analytic), Latin (inflectional), synthetic n-gram text, and JSON-like structured data. All datasets are size-matched to the Voynich corpus to ensure fair comparison. The Latin corpus was obtained from Project Gutenberg (ID 18837, Caesar, Commentarii de Bello Gallico, Books V–VIII). Preprocessing included lowercasing,

removal of non-alphabetic characters ([^a–z] → space), whitespace normalization, and tokenization based on whitespace segmentation. All baseline corpora were processed using identical normalization procedures to ensure cross-dataset comparability.

4.2 Metrics

The study employs entropy decay, ambiguity persistence (candidate survival rate), and prefix-family distributions. Prefix and suffix are defined operationally as the first and last n-character segments of a token, respectively. This definition enables consistent comparison across datasets without assuming an underlying linguistic or morphological structure.

4.3 Null Models and Statistical Testing

Three null models were constructed: frequency-preserving shuffle, length-preserving randomization, and a prefix-preserving model with a global suffix distribution. Permutation-based significance testing (N = 1,000) was applied to all correlation scores.

5. Results

5.1 Cross-Dataset Comparison

Voynich text shows the highest similarity to natural language, with entropy and ambiguity correlations exceeding 0.95 ($p < 0.01$). Notably, the structured baseline (JSON) exhibits a pronounced entropy spike at intermediate prefix lengths, reflecting the transition from predictable syntactic templates to unpredictable data values. This "data spike" is absent in both Voynich and natural language, which follow a smooth decay.

Dataset	Type	Similarity	Entropy Corr.	Ambiguity Corr.
english.txt	natural (analytic)	0.5502 ± 0.0018	0.9500 ± 0.0004	0.9522 ± 0.0048
latin.txt	natural (inflectional)	0.5156 ± 0.0021	0.9849 ± 0.0003	0.9373 ± 0.0041
ngram.txt	natural (synthetic)	0.4618 ± 0.0000	0.9134 ± 0.0000	0.7198 ± 0.0000
json.txt	structured data	0.2725 ± 0.0008	–0.3143 ± 0.0045	–0.3475 ± 0.0062

Table 1. Size-matched cross-dataset similarity (N = 39,049). Similarity scores reflect the mean Pearson correlation of entropy decay curves across bootstrap samples. Lower std indicates stronger robustness across size-matched resamples.

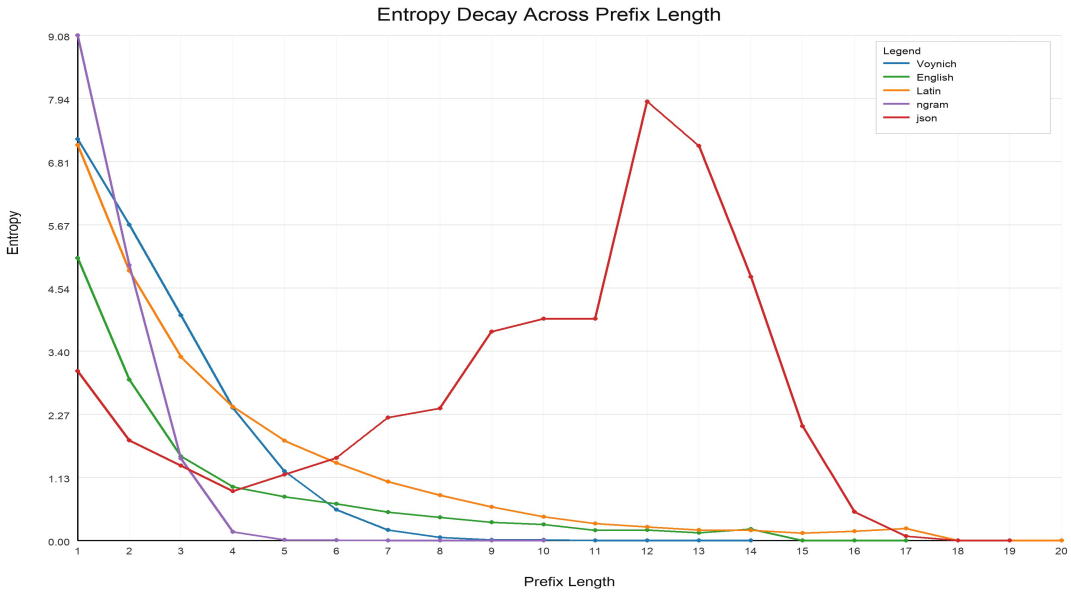

Figure 1. Entropy decay curves across prefix length. Voynich follows a natural-language-like trajectory more closely than synthetic or structured baselines.

5.2 Structural Divergence

Despite statistical similarity, structural comparisons reveal substantial differences. Prefix-family overlap between Voynich and natural language baselines remains minimal (2–3%). Additionally, Voynich tokens exhibit highly concentrated suffix distributions, indicating a rigid positional organization that differs from natural language morphology.

Null Model	Entropy Diff (L2)	Candidate Diff (L2)	Ambiguity Diff (L2)	Prefix Overlap
frequency_shuffle	4.5191	570.49	1.0534	2.56%
prefix_global_suffix	4.6961	823.58	1.3786	48.15%
length_preserving	4.4048	1528.20	1.0455	0.00%

Table 2. Divergence of null models from real Voynich data.

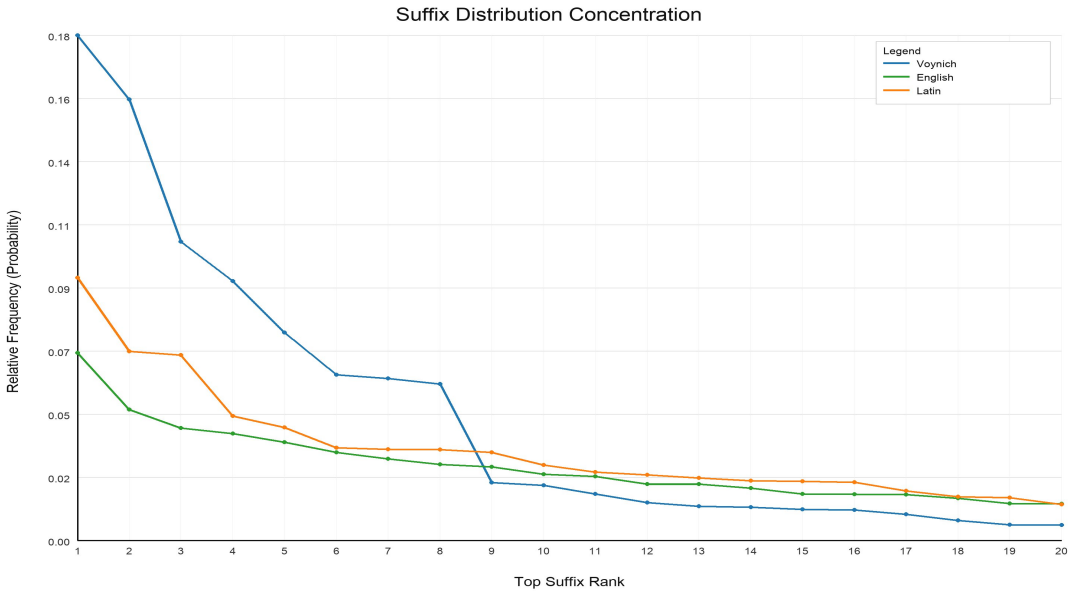

Figure 2. Ranked suffix distributions show that Voynich suffix usage is more concentrated than natural-language baselines, indicating structural divergence despite similar predictability dynamics.

5.3 Null Model Evaluation

Null models fail to reproduce the observed patterns, showing a rapid collapse of ambiguity beyond prefix length 4. Voynich tokens maintain ambiguity across longer segments, suggesting the presence of deep, higher-order dependencies that cannot be explained by frequency, length, or simple prefix-suffix sampling.

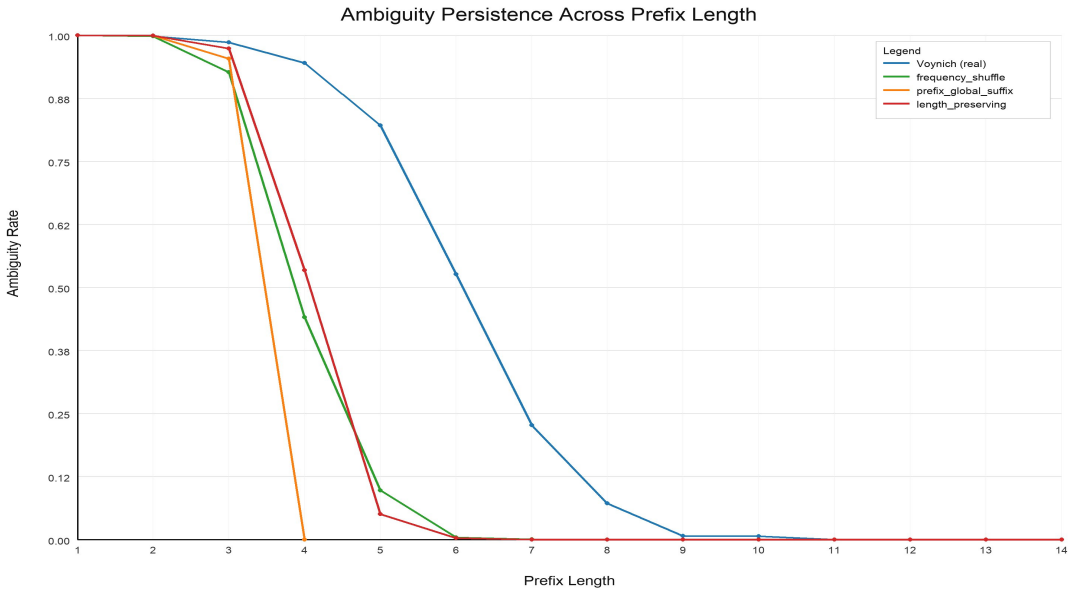

Figure 3. Ambiguity persistence by prefix length. All three null models collapse to near-zero ambiguity by prefix length 6, while Voynich maintains substantial ambiguity through length 8. This gap, reproduced across frequency-preserving, length-preserving, and prefix-constrained baselines, constitutes direct evidence for higher-order generative dependencies that

cannot be reduced to any combination of token frequency, length distribution, or prefix-suffix co-occurrence.

5.4 Research Question Mapping

RQ1: Voynich aligns statistically with natural language across both analytic and inflectional types ($\rho \approx 0.95$, $p < 0.01$).

RQ2: Simple null models fail to replicate the observed ambiguity persistence.

RQ3: Structural organization diverges substantially from natural language, with Voynich showing greater suffix concentration and minimal prefix-family overlap.

RQ4: Language-like predictability and structural constraint coexist in the Voynich system, consistent with a hybrid generative framework.

6. Discussion

The findings suggest a hybrid system in which language-like predictability coexists with strong positional and morphological constraints. This statistical profile does not correspond directly to known encoding frameworks but rejects the hypothesis of a trivial hoax or random text. The consistency across English and Latin indicates that the observed similarity is a robust property of the Voynich system itself. The use of a Latin corpus with relatively regular narrative structure (Caesar's Commentarii) may contribute to the high entropy correlation observed ($\rho = 0.9849$), suggesting that the Voynich predictability profile aligns with structured narrative text rather than free-form prose.

The persistence of ambiguity across intermediate prefix lengths — a property not reproduced by any tested null model — suggests that Voynich tokens encode higher-order dependencies that are destroyed by even frequency-preserving randomization. Whether these dependencies reflect grammatical structure, an encoding scheme, or a novel generative system remains an open question.

These results complement and extend earlier findings on positional token constraints (Chang, 2026a), which demonstrated that specific token classes are statistically bound to line-initial, paragraph-initial, and line-final positions. Together, the two studies suggest that Voynich text is governed by both intra-token generative rules and inter-position structural constraints.

7. Limitations

This study focuses exclusively on intra-token structure. It does not model inter-token syntactic relationships or long-range dependencies across paragraphs, which remains a direction for future research. The baseline datasets, while carefully selected, may not capture the full range of symbolic systems relevant to the Voynich question.

8. Conclusion

Voynich text demonstrates robust statistical similarity to natural language alongside clear structural divergence. Entropy and ambiguity correlations exceeding 0.95 against both analytic and inflectional language baselines provide quantitative evidence that the manuscript's token system shares deep generative properties with natural language. At the same time, minimal prefix-family overlap, concentrated suffix distributions, and ambiguity persistence patterns that no tested null model can

reproduce strongly support the interpretation of a structured symbolic system governed by unique generative rules. These findings reject both the trivial hoax hypothesis and the hypothesis of direct natural language encoding. The convergence of intra-token predictability structure with previously established positional constraints (Chang, 2026a) indicates that Voynich text is organized at multiple structural levels simultaneously — a property that any future generative account of the manuscript must explain.

Acknowledgements

AI-based tools (Claude, ChatGPT, Gemini) assisted with code implementation, figure generation, and manuscript editing. Research design, analytical decisions, and interpretation were conducted by the author. The reproduction package for Chang (2026a) is available at https://doi.org/10.5281/zenodo.19689754.

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